

ABSTRACT

Title: Hospital Information and Patient Management System

Objective:

Hospital Information and Patient Management System is a web-based application designed to provide a structured, secure, and intelligent platform for managing all core operations of a hospital or healthcare facility. The main aim of this project is to build a centralised digital system that eliminates the need for manual paper-based record keeping and disconnected software tools by bringing patient management, doctor scheduling, appointment booking, billing, and medical records all under one unified platform. The system also integrates an Artificial Intelligence module that uses machine learning algorithms to predict patient readmission risk and assist doctors in early diagnosis based on reported symptoms.

In most hospitals today, especially small and mid-sized facilities, administrative tasks such as registering patients, scheduling appointments, tracking bed availability, generating bills, and maintaining medical histories are still handled manually or through outdated systems. This leads to errors, delays, data loss, and a poor experience for both patients and healthcare staff. The proposed system addresses all of these challenges by providing a modern, fully digital platform that automates routine tasks, reduces human error, improves the speed and quality of patient care, and gives hospital administrators complete visibility and control over all hospital operations through a real-time dashboard.

The system is designed to serve three categories of users the hospital administrator, doctors, and patients each with their own secure login, dedicated dashboard, and specific set of features tailored to their role. By combining a robust Django backend with a clean and responsive frontend and a machine learning prediction engine, this project demonstrates how technology can be used to improve healthcare delivery in a meaningful and practical way.

Existing Systems:

Currently, most small and medium-sized hospitals manage their day-to-day operations through a combination of paper registers, spreadsheets, and disconnected software tools that do not communicate with each other. Larger hospitals may use expensive proprietary Hospital Management Systems, but these are often too costly for smaller facilities and are difficult to customise. The key limitations of existing systems include:

- There is no centralised platform that connects patient records, doctor schedules, appointment booking, billing, and reporting in one place, forcing staff to switch between multiple tools or maintain duplicate records.
- Patient registration and appointment booking are handled manually, leading to long waiting times, scheduling conflicts, double bookings, and a frustrating experience for patients.
- Medical histories and previous prescriptions are stored in physical files that can be easily lost, damaged, or misplaced, with no reliable way to retrieve them quickly in an emergency.

- There is no real-time visibility into bed availability, doctor availability, or ward occupancy, making it difficult for administrators to manage resources efficiently during busy periods.
- Billing and invoice generation are done manually, which is time-consuming, error-prone, and makes it difficult to maintain accurate financial records.
- Existing systems have no integration with Artificial Intelligence or machine learning, meaning doctors receive no automated support for early diagnosis, risk prediction, or treatment recommendations.
- There is no mechanism for patients to access their own medical records, appointment history, prescriptions, or test reports online, forcing them to visit the hospital physically for even routine information.
- Reports and analytics are generated manually and are often outdated, giving management very little insight into hospital performance, patient trends, or resource utilisation.

Proposed System:

The Hospital Information and Patient Management System addresses all the limitations of existing systems by providing a comprehensive, intelligent, and fully digital platform for hospital management. The proposed system includes the following features:

- A centralised web platform built using the Django framework that manages all hospital operations including patient registration, doctor management, appointment scheduling, medical records, ward management, billing, and reporting from a single unified interface.
- A secure role-based authentication system that provides separate login portals and dashboards for administrators, doctors, and patients, ensuring that each user can only access information and features relevant to their role.
- A complete patient management module that allows staff to register new patients, maintain detailed profiles including personal information, medical history, allergies, and previous prescriptions, and retrieve records instantly by searching patient ID or name.
- An online appointment booking system that allows patients to browse available doctors by specialisation, check real-time availability, book appointments, receive confirmation, and cancel or reschedule if needed — all without having to visit or call the hospital.
- A doctor management module that allows administrators to manage doctor profiles, assign specialisations, set schedules and consultation hours, track patient load, and monitor daily appointments through a dedicated dashboard.
- A real-time bed and ward management system that tracks the availability and occupancy status of all beds across different wards, allowing administrators to assign beds to admitted patients and monitor capacity at a glance.

- An automated billing and invoice generation module that calculates charges based on consultation fees, procedures, medications, and bed charges, generates itemised invoices, and maintains a complete financial ledger for each patient.
- An Artificial Intelligence module powered by machine learning that includes a patient readmission risk predictor built using XGBoost and a symptom-based disease prediction tool built using a Random Forest classifier, giving doctors an intelligent second opinion to support their clinical decisions.
- A comprehensive reporting and analytics dashboard that displays real-time statistics on patient registrations, appointments, admissions, revenue, doctor performance, and disease trends using interactive charts and graphs.
- A patient portal that allows registered patients to log in and view their own appointment history, medical records, prescriptions, test reports, and bills from anywhere without needing to visit the hospital.

Actual Users of the System:

1. Admin (Hospital Administrator)

- Login to the system using secure admin credentials.
- Register and manage doctor profiles, specialisations, and schedules.
- Register new patients and manage their complete profiles and medical records.
- Monitor real-time bed availability and assign wards to admitted patients.
- Generate and manage patient bills and invoices.
- View hospital-wide analytics and performance reports through the admin dashboard.
- Manage system settings including departments, wards, and user accounts.
- Log out from the system.

2. Doctor

- Login to the personal doctor dashboard using secure credentials.
- View the list of appointments scheduled for the day and upcoming days.
- Access complete patient history, previous prescriptions, and medical records before consultation.
- Record consultation notes, diagnoses, and prescriptions after each appointment.
- View AI-generated readmission risk scores and symptom-based disease predictions for patients.
- Request lab tests and view test results linked to the patient record.
- Log out from the system.

3. Patient

- Register for an account and login to the patient portal.
- Browse doctors by specialisation and check real-time appointment availability.
- Book, cancel, or reschedule appointments online.
- View personal medical history, previous prescriptions, and diagnoses.
- Download or view itemised bills and payment receipts.
- View lab test results and doctor notes uploaded after consultation.
- Log out from the portal.

Technology Stack:

- Programming Language: Python
- Backend Web Framework: Django
- Frontend: HTML, CSS, Bootstrap, JavaScript
- Database: MySQL
- Machine Learning: Scikit-learn (Random Forest, XGBoost)
- Data Processing: Pandas, NumPy
- Charts and Visualisation: Chart.js
- Authentication: Django Auth with Role-Based Access Control